TELIPORT
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Season 4

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## Case Study 2

# Team StratWise

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THINK. BUILD. CREATE. WIN

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# Test a battery before you build it

VOLT-FORGE connects your battery design decisions to what they physically change

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PROBLEM UNDERSTANDING

## Cells rarely share the load evenly

Say a Raspberry Pi 5 gateway must survive a **10-minute power cut at 25 W**, its full supply rating, at 23 °C. Four cells in parallel sounds easy, until the cell closest to the output takes more current at first. Ageing and packaging come later.

DESIRABILITY

## You shouldn't need a lab to learn that

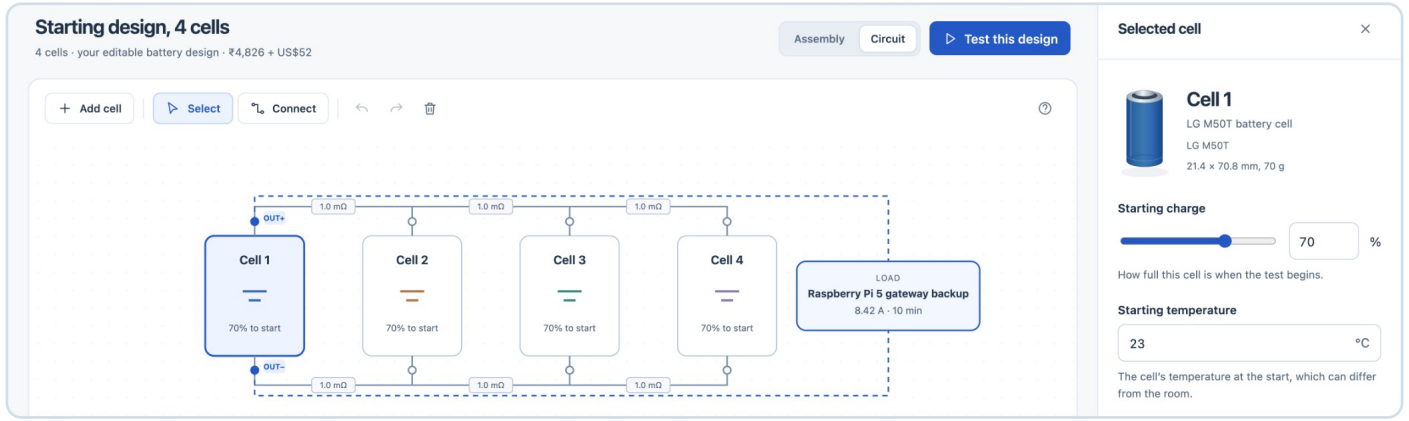
Today that takes a bench, a paid tool or your own code. Students can't afford that and engineers can't wait. Here you wire the circuit on screen and see which cell struggles **before spending a rupee**.

TECHNICAL FEASIBILITY

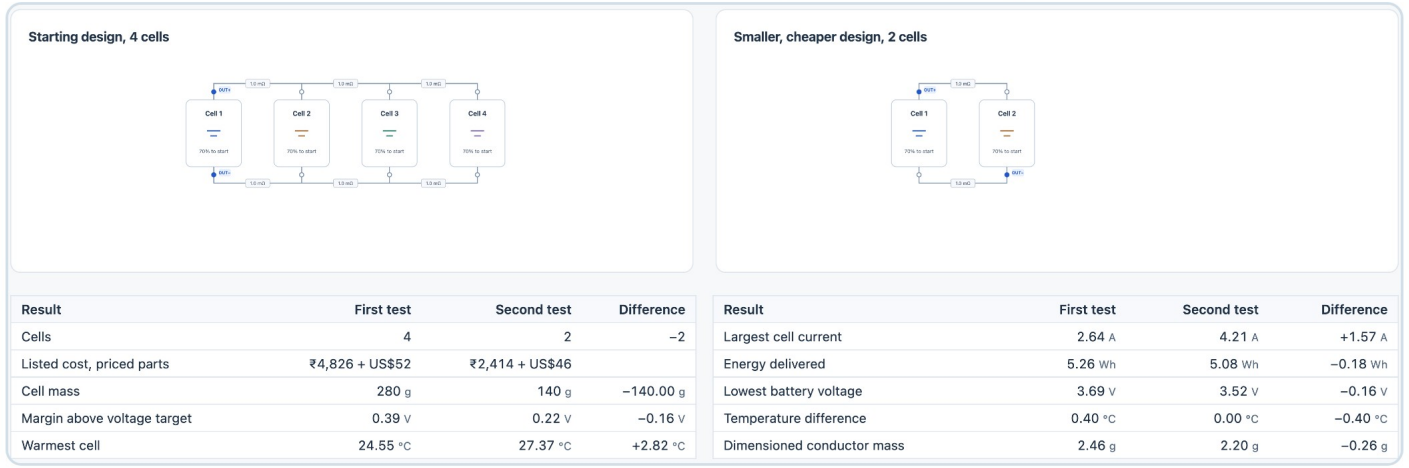
## So how few cells can do this job?

We compared 3 and 2 cells against the 4-cell start. **Two cells** held the Pi for all 10 minutes in all 33 simulated tests, even with one weak cell. One cell was refused. It should fit a two-cell Pi UPS board, though we haven't built it yet, and the tool runs on a laptop.

PROTOTYPE & EVIDENCE Test it out: **volt-forge-app.vercel.app**




**Build it.** This is the real circuit the solver runs. Click any cell to see and change what's inside it.




| Result | First test | Second test | Difference |
|---|---|---|---|
| Cells | 4 | 2 | −2 |
| Listed cost, priced parts | ₹4,826 + US$52 | ₹2,414 + US$46 | |
| Cell mass | 280 g | 140 g | −140.00 g |
| Margin above voltage target | 0.39 V | 0.22 V | −0.16 V |
| Warmest cell | 24.55 °C | 27.37 °C | +2.82 °C |

| Result | First test | Second test | Difference |
|---|---|---|---|
| Largest cell current | 2.64 A | 4.21 A | +1.57 A |
| Energy delivered | 5.26 Wh | 5.08 Wh | −0.18 Wh |
| Lowest battery voltage | 3.69 V | 3.52 V | −0.16 V |
| Temperature difference | 0.40 °C | 0.00 °C | −0.40 °C |
| Dimensioned conductor mass | 2.46 g | 2.20 g | −0.26 g |

**Compare it.** Same 25 W job: 2 cells cost and weigh half, but work harder with less voltage margin.

**4,006 simulated cases.** 13 designs against weak cells, bad links and poor cooling. A full rerun matched exactly.

**54 published four-cell experiments.** All 194 dataset files verified intact, 2 tests compared with the model so far.

**9 records kept closed.** Saved for a final exam. 2 can test this cell, both at 40 °C.

Every cost, source and check is in the full appendix: **volt-forge-app.vercel.app/appendix**

ENGINEERING QUALITY

## Why should you trust it?

Every run must balance its energy, which catches solver errors, not a wrong model. Against published data, voltage and temperature missed all 12 checks. Current sharing came closer but also missed: the nearest cell took **34%** early on, the model said **34%**. Over 10 minutes, real heating stayed within 1.13 times the model.

INNOVATION

## You don't have to guess

Press **Find the smallest design**. It rebuilds the circuit at each size and tests it under normal conditions. For the Pi: 2 cells, not 4, at half the cell cost. Our study then stress-tested that.

BUSINESS VIABILITY



**41.3 lakh** TAM, engineering students

**15.6 lakh** SAM, electrical, electronics, mechanical

**130** SOM, labs at ₹50,000, 5% assumed

## ₹35 crore

to build it all. Stage one is ₹11 crore: 130 labs and 31 teams at ₹5 lakh a year would repay it in 5 years, before running costs. Students use it free. No customer has seen these prices.

**Simulated** with an assumed 90% converter and 3.3 V floor. **Next:** a bench test with voltage, current and heat cutoffs, then student and engineer trials.